Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI-Powered Agents · Python · C# · Java · Agentic Systems · Production AI · Observability · CI/CD

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SUMMARY

Senior Software Engineer with 8+ years owning production systems end-to-end - from early design through deployment, production operations, and ongoing reliability improvement. Built **Leasel**, an 8-agent LangGraph production system with durable state, conditional routing, trace-level observability, audit logging, and a CI regression harness: designed, implemented, validated, and owns it in production. Built **AgentProbe**, an open-source AI evaluation toolkit with CI-integrated regression testing and LLM-as-judge scoring - used daily as an AI coding tool. At Entrata (8 years, 100K+ monthly active users), owned complex production systems across the full SDLC: requirements, design, implementation, testing, deployment, production incident resolution, and mentoring. Strong Python, C# (.NET, 1.5yr), and Java; experienced with Azure AI services, MCP, agentic workflows, and responsible AI patterns (HITL, guardrails, audit trails). M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

AI-Powered Agents · Agentic Workflows · Python · C# · Java · JavaScript · LangGraph · LangChain · MCP · Tool Use · RAG · Hybrid Search · Azure AI Services · AWS · GCP · Observability · Tracing · CI/CD · GitHub Actions · Responsible AI · HITL · Guardrails · Production Operations · Distributed Systems · REST APIs · OpenAPI · PostgreSQL · Redis · SQLite · Docker · Kubernetes · Security & Privacy Engineering · Mentorship · Code Review

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models and AI-powered features end-to-end - data pipelines, backend APIs, and integration layers; applying responsible AI practices and debugging across the full stack.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built an agentic mission-planning system (LangGraph StateGraph) with tool use, RAG over regulatory documents, validation guardrails, and conditional routing - owned from prototype to deployed integration.
- Deployed an **MCP (Model Context Protocol) server** connecting agentic pipeline outputs to downstream enterprise tooling: authentication, structured access, failure routing, and observability across the integration boundary.
- Engineered full document ingestion, chunking, retrieval, and FastAPI streaming pipeline with defined API contracts and retry semantics.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, multi-tenant)

- **Owned complex production systems across the full SDLC** - requirements analysis, design, implementation, testing, deployment, production monitoring, and incident resolution for 7+ years in a regulated environment serving 100K+ users; zero compliance violations.
- **Led production incident resolution across services and multiple stack layers** - built internal debugging tools that reduced debugging time by 40%; improved observability and resiliency practices to prevent recurrence; maintained bug count below 10 for 6+ consecutive months.
- **Drove engineering excellence and developer productivity** through 300+ peer code reviews, design reviews, automated test pipelines, and CI/CD automation - Python/Bash automation pipelines eliminated 80% of manual operational workload and accelerated team delivery velocity; led microservices redesign (Docker) achieving 50% scalability improvement and 30% deployment time reduction.
- Owned distributed async processing (RabbitMQ), data pipeline (federal rows → 180K domain records), AWS S3 migration with PHI compliance (Libsodium encryption, zero security incidents), and complex multi-tenant PostgreSQL schema optimization (200+ queries, 30% gain).
- Mentored 5+ engineers and led cross-team technical decisions; shipped 20+ full-stack modules (Python, PHP, C#-adjacent .NET patterns, Java, JavaScript) across 120+ Agile sprints.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration from legacy storage; delivered compliance data workflows under senior engineer guidance.

PROJECTS

Leasel - Production Agentic AI System 2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Designed, implemented, validated, and operates in production**: 8-agent LangGraph system with durable state, tool use, conditional routing, trace-level observability, structured audit logging, and HITL escalation - responsible AI patterns enforced in code throughout.
- CI regression harness with LLM-as-judge scoring catches agent degradations before they reach users; 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

AgentProbe - Open-Source AI Evaluation Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; multi-provider LLM routing (Anthropic, Azure OpenAI, OpenAI, Groq, Ollama); 5-metric eval engine; CI regression harness with golden datasets - AI-assisted development tool used daily to improve code quality and catch regressions.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017